Criterion 5: Faculty Information (100)

by
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Faculty details

S.N.	Name of Faculty	PAN No.	APAAR Faculty ID (if any)	Highest Degree	University	Area of Specialization	Date of Joining in Institution	Experience in Current Institute (Years)	Designation at Joining	Present Designation	Date of Promotion (Assoc./Prof.)	Nature of Association (Regular/Contract/Adhoc)	If Contractual – Type (Full-time/Part-time/Hourly)	Currently Associated (Y/N)	Date of Leaving (if applicable)	Remarks / Evidence (Appointment, Payroll, Approval)
1	Dr. A.B. Expert	XXXXX1234X	APAAR001	Ph.D. (CSE)	IIT Delhi	Artificial Intelligence	01-Jul-2018	6.0	Assistant Professor	Associate Professor	01-Jan-2022	Regular	–	Y	–	Appointment + Approval + Payroll
2	Prof. C.D. Mentor	YYYYY5678Y	APAAR002	Ph.D. (EEE)	NIT Trichy	Power Systems	12-Aug-2019	5.0	Associate Professor	Professor	15-Mar-2023	Regular	–	Y	–	Appointment + Promotion Order
3	Ms. E.F. Scholar	ZZZZZ9876Z	APAAR003	M.Tech. (Data Science)	Anna University	Data Analytics	10-Sep-2022	1.8	Assistant Professor	Assistant Professor	–	Contract	Full-time	Y	–	Contract Letter + Payroll
4	Dr. G.H. Senior	PPPPP4321P	–	Ph.D. (ME)	IISc Bangalore	Thermal Engineering	20-Jul-2020	3.8	Assistant Professor	Assistant Professor	–	Regular	–	N	30-Jun-2024	Relieving Order Attached

5.1. Student-Faculty Ratio (30)

- **Purpose of SFR**
  - Evaluates whether a department has an adequate number of faculty members relative to its student strength.
  - Ensures quality of teaching, mentoring, and research guidance.
  - A lower ratio (fewer students per faculty) reflects better academic standards.
- **Scope of Calculation**
  - Calculated at the Department level.
  - Must include:
    - All Undergraduate (UG) programs under the department and allied clusters.
    - All Postgraduate (PG) programs under the department and allied clusters.
  - Faculty engaged exclusively in First Year teaching are excluded.
- **UG Program Contribution**
  - For each UG program (UG1 … UGn):
    - B = No. of students in 2nd year.
    - C = No. of students in 3rd year.
    - D = No. of students in 4th year.
  - First year students are excluded because their faculty is also excluded from count.

- **PG Program Contribution**
  - **For each PG program (PG1 … PGm):**
    - **A = No. of students in 1st year.**
    - **B = No. of students in 2nd year.**
- **Student Count (S)**
  - **S = Total eligible students of UG (2nd–4th year) + PG (1st–2nd year) across all programs.**
  - **Student strength (ST) is computed as:**
    ST = Sanctioned Intake (SA) + Lateral Entry students (L up to 10% of SA)
  - **Exclusions from S:**
    - **Students admitted under supernumerary quotas (SNQ, EWS, TFWS, etc.).**
- **Faculty Count (F)**
  - **Includes:**
    - **Full-time regular faculty.**
    - **Full-time contractual faculty (with full workload).**
    - **Faculty from allied departments/clusters contributing to the program.**
  - **Excludes:**
    - **Faculty teaching only First Year courses (100% workload in 1st year).**

- **Formula for SFR**
  - **SFR = S ÷ F**
  - **Where:**
    - **S = Total eligible student strength.**

    F = Total eligible faculty strength

- **Key Documentation for Audit**
  - **Student intake approvals (AICTE/University sanctioned seats).**
  - **Admission registers (showing 2nd year lateral entry, excluding SNQ/EWS).**
  - **Faculty appointment letters (full-time/contractual proof).**
  - **Workload allocation records (to justify exclusion of 1st-year faculty).**
  - **Supporting evidence from allied departments (if faculty is shared).**

calculate the number of students admitted via lateral entry or leftover seats for NBA SFR calculation using your example:

1. Parameters

- SA = Sanctioned Intake in 1st Year (e.g., 120)
- L = Students admitted in 2nd year via
  - a = Lateral entry (max 10% of SA)
  - b = Leftover seats (vacant in 1st year, admitted in 2nd year)
- ST = Total students considered for SFR = SA + L
- Constraint: L ≤ 10% of SA; ST ≤ SA × 1.1 (capped at 132 in example)

2. Rules for Calculation

- Case 1: L ≤ 10% of SA → ST = SA + L
- Case 2: L > 10% of SA → ST = SA + 10% of SA (cap)
- Maximum ST allowed = SA × 1.1
- Any leftover or lateral entry above 10% limit is ignored.

Example Cases

Case	SA	1st Yr Admitted	Leftover Seats	L = a+b (2nd Yr)	ST = SA + min(L, 10% SA)	Explanation
1	120	120	0	0	120 + 0 = 120	No extra students
2	120	120	0	12	120 + 12 = 132 (Cap)	Full 10% lateral entry allowed
3	120	120	0	6	120 + 6 = 126	6 < 10%, so added fully
4	120	60	60	0	120 + 0 = 120	No lateral, leftover admitted only
5	120	75	45	6	120 + 6 = 126	L=6 < 12, so allowed fully
6	120	82	38	12	120 + 12 = 132 (Cap)	L=12 =10%, max allowed
7	120	88	32	44	120 + 12 = 132 (Cap)	L=44 >10%, only 10% allowed
8	120	60	60	42	120 + 12 = 132 (Cap)	L=42 >10%, only 10% allowed

Student–Faculty Ratio (Department + Allied)

Item	CAY	CAYm1	CAYm2
UG1.B (2nd Year Students)			
UG1.C (3rd Year Students)			
UG1.D (4th Year Students)			
UG1 = B+C+D			
… repeat for UG2, UG3 … UGn			
PG1.A (1st Year Students)			
PG1.B (2nd Year Students)			
PG1 = A+B			
… repeat for PG2 … PGm			
DS (Department Students)			
AS (Allied Students)			
S = DS + AS (Total Students)	S1	S2	S3
DF (Department Faculty)			
AF (Allied Faculty)			
F = DF + AF (Total Faculty)	F1	F2	F3
FF (First-Year Only Faculty)	FF1	FF2	FF3
Effective Faculty = F – FF			
SFR = S / (F – FF)	SFR1	SFR2	SFR3
Average SFR = (SFR1+SFR2+SFR3)/3			

How to fill this table

- UGi.B/C/D: Enter ST_final values (after applying SA+L caps, excluding SNQ).
- PGj.A/B: Same rule for PG programs.
- DS: Sum of all UG + PG students in department.
- AS: Students from allied clusters serving your programs.
- S = DS + AS.
- DF, AF: Count full-time faculty (excluding first-year-only).
- FF: Faculty with 100% first-year load.
- SFRy = Sy ÷ (Fy−FFy).
- Average SFR: arithmetic mean of SFR1, SFR2, SFR3.

Marks calculation logic for NBA SFR

1. Context

- NBA evaluates Student–Faculty Ratio (SFR) over 3 years (CAY, CAYm1, CAYm2).
- After computing SFR1, SFR2, SFR3, we take the Average SFR = (SFR1 + SFR2 + SFR3) / 3.
- Marks are then awarded out of 30 based on this Average SFR.

2. Marking Criteria (as per your note)

- SFR < 15 → 30 Marks (full marks).
- 15 ≤ SFR < 17 → 26 Marks.
- 17 ≤ SFR < 19 → 22 Marks.
- 19 ≤ SFR < 21 → 18 Marks.
- 21 ≤ SFR < 23 → 14 Marks.
- 23 ≤ SFR < 25 → 10 Marks.
- SFR ≥ 25 → 0 Marks (no marks).

3. Mathematical Expression

- The rule between SFR = 15 and 25 is linear with slope −2 marks per +1 in SFR.
- Formula:

  $$\text{Marks}=30-2\times(\text{SFR}-15)$$

- Boundaries:
  - If SFR < 15 → cap at 30.
  - If SFR > 25 → floor at 0.

Example Mapping

Average SFR	Marks Awarded
14.5	30
16.0	26
18.0	22
20.0	18
22.0	14
24.0	10
26.0	0

Computer Science & Engineering (CSE) Department (Cluster Programs) as per your example

1. Concept of Clustering

- **NBA groups multiple specializations under one parent department as a cluster.**
- **For CSE, the following UG & PG programs are treated as one single CSE Department cluster:**
  - **UG Programs (6):**
    - **UG1 = B.Tech CSE**
    - **UG2 = B.Tech CSE (Artificial Intelligence)**
    - **UG3 = B.Tech CSE (AI & Machine Learning)**
    - **UG4 = B.Tech CSE (Cyber Security)**
    - **UG5 = B.Tech CSE (Data Science)**
    - **UG6 = B.Tech Information Technology *(considered Allied)***
  - **PG Programs (1):**
    - **PG1 = M.Tech CSE**
- **Key Rule: When a cluster is formed, all students and faculty in those programs must be considered together for SFR .**

2. Student Calculation (S)

- **UG Programs (UG1 to UG5) = Department students (DS).**
- **UG6 (Information Technology) = Allied Department students (AS).**
- **PG1 (CSE) = Department students.**
- **Final Student Count (S):**
  - **S = DS + AS**
  - **Example:**
    - **CAY: DS = 1,964, AS = 393 → S1 = 2,357**
    - **CAYm1: DS = 1,954, AS = 391 → S2 = 2,345**
    - **CAYm2: DS = 1,919, AS = 388 → S3 = 2,307**

3. Faculty Calculation (F)

- DF (Department Faculty): All full-time faculty in CSE programs.
- AF (Allied Faculty): Faculty from IT or allied departments teaching in the cluster.
- F = DF + AF
- FF: Faculty with 100% first-year teaching load (excluded).
  - In this case, FF = 0 (no such faculty).
- Final Faculty Counts:
  - CAY: DF = 95, AF = 25 → F1 = 120
  - CAYm1: DF = 99, AF = 25 → F2 = 124
  - CAYm2: DF = 100, AF = 25 → F3 = 125

4. SFR Formula

SFR=S/(F−FF)

- CAY (SFR1):
  - S1 = 2,357; F1 = 120; FF1 = 0
  - SFR1 = 2,357 ÷ 120 = 19.64
- CAYm1 (SFR2):
  - S2 = 2,345; F2 = 124; FF2 = 0
  - SFR2 = 2,345 ÷ 124 = 18.91
- CAYm2 (SFR3):
  - S3 = 2,307; F3 = 125; FF3 = 0
  - SFR3 = 2,307 ÷ 125 = 18.46

5. Average SFR (3 years)

Average SFR=(SFR1+SFR2+SFR3)/3

=(19.64+18.91+18.46)/3=19.00

6. Marks Allocation (Based on NBA Rules)

- **From your marking scale:**
  - **SFR < 15 → 30 Marks**
  - **15–25 → proportional (linear decrease)**
  - **> 25 → 0 Marks**
- **Since Average SFR = 19.00, it falls in the 15–25 range.**
  - **Marks = 30−2×(19−15)=22**

7. Key Observations

- **Clustering Impact: Even though multiple UG programs exist, they are not evaluated separately; NBA ensures faculty adequacy across all specializations collectively.**
- **Allied Programs (IT): Their students & faculty must be included in the SFR if the department/cluster teaching overlaps.**
- **Balanced Faculty Load: Since FF = 0, all faculty are counted; if some were teaching only 1st year, they would be excluded, worsening SFR.**
- **Final NBA Evaluation: Department's Average SFR = 19.00 → 22 Marks out of 30 for Criterion 5.1.2.**

Student–Faculty Ratio (CSE Cluster Programs)

Item	CAY	CAYm1	CAYm2
UG1.B (CSE – 2nd Year)			
UG1.C (CSE – 3rd Year)			
UG1.D (CSE – 4th Year)			
UG1 = B + C + D			
UG2.B (CSE–AI – 2nd Year)			
UG2.C (CSE–AI – 3rd Year)			
UG2.D (CSE–AI – 4th Year)			
UG2 = B + C + D			
UG3.B (CSE–AIML – 2nd Year)			
UG3.C (CSE–AIML – 3rd Year)			
UG3.D (CSE–AIML – 4th Year)			
UG3 = B + C + D			
UG4.B (CSE–CyberSec – 2nd Year)			
UG4.C (CSE–CyberSec – 3rd Year)			
UG4.D (CSE–CyberSec – 4th Year)			
UG4 = B + C + D			
UG5.B (CSE–Data Science – 2nd Year)			
UG5.C (CSE–Data Science – 3rd Year)			
UG5.D (CSE–Data Science – 4th Year)			
UG5 = B + C + D			
UG6.B (Allied IT – 2nd Year)			
UG6.C (Allied IT – 3rd Year)			
UG6.D (Allied IT – 4th Year)			
UG6 = B + C + D (Allied)			
PG1.A (CSE – 1st Year)			
PG1.B (CSE – 2nd Year)			
PG1 = A + B			
DS = Dept Students (UG1..UG5 + PG1)			
AS = Allied Students (UG6)			
S = DS + AS (Total Students)			
DF = Dept Faculty			
AF = Allied Faculty			
F = DF + AF			
FF = First-Year-only Faculty			
Effective Faculty = F – FF			
SFR = S / (F – FF)			
Average SFR = (SFR1+SFR2+SFR3)/3			
Marks (out of 30)			

5.2. Faculty Qualification (25)

1. Purpose of this Criterion

- This sub-criterion evaluates the academic quality of faculty in a department/cluster.
- It rewards departments that maintain a good proportion of Ph.D.-qualified and M.Tech/ME-qualified faculty.
- The metric used is the Faculty Qualification Index (FQI), which links faculty qualifications with required strength (RF).

2. Formula for Faculty Qualification Index (FQI)

$$FQI=2.5\times(10X+4Y)/RF$$

- X = Number of faculty members with Ph.D. (or equivalent) degree.
- Y = Number of faculty members with M.Tech/ME (or equivalent) degree.
- RF = Required Faculty = S/20
  - Where S = Number of students in department (including allied programs).

    20:1 is the NBA-mandated Student–Faculty Ratio

3. Meaning of Each Term

- **10X → Each Ph.D. faculty gets 10 points weight.**
- **4Y → Each M.Tech faculty gets 4 points weight.**
- **RF → Denominator ensures faculty quality is measured relative to the faculty required (not just those available).**
- **2.5 multiplier → Normalization to bring maximum score to 25 marks.**

4. Step-by-Step Calculation Procedure

1. **Count students (S):**
   - **From approved admissions (SA + lateral entries ≤110% SA, excluding SNQ/EWS).**
   - **Include students of all UG & PG programs in department + allied departments.**
2. **Calculate RF:**

   RF=S/20

   Example: If total students S=600S = 600S=600, then RF=600/20=30
3. **Count faculty:**
   - **X = Faculty with Ph.D. (approved, full-time, counted in SFR).**
   - **Y = Faculty with M.Tech/ME.**
   - **(Exclude first-year only faculty (FF), visiting, guest, part-time).**
4. **Compute Weighted Faculty Value:**

   10X+4Y
5. **Apply Formula:**

   FQI=2.5×(10X+4Y)/RF
6. **Cap maximum:**
   - **If FQI > 25, award full 25 marks only.**

5. Example Calculation

- Total students (S) = 600
- Required Faculty (RF) = 600/20 = 30
- Faculty with Ph.D. (X) = 12
- Faculty with M.Tech (Y) = 18

Step 1: Weighted faculty value = (10×12 + 4×18) = 120 + 72 = 192
Step 2: Divide by RF = 192/30 = 6.4
Step 3: Multiply by 2.5 = 16.0

FQI = 16.0 marks (out of 25)

6. Key NBA Compliance Points

- Faculty considered must be AICTE/UGC approved and full-time.
- Allied department faculty are included in X, Y, and RF.
- If RF is higher than available faculty, it still applies (to prevent artificially inflated FQI).

First-year faculty (100% load) excluded from X & Y

7. Documentation Required

- Faculty Qualification Proofs
  - Ph.D./M.Tech degree certificates.
  - University/AICTE faculty approvals.
- Faculty List (mapped with qualification, designation, DOJ, dept).
- Student Strength Records (admissions, registers) to justify S.
- Cross-verification with SFR records (Criterion 5.1).

8. Marking

- FQI directly converts to marks (capped at 25).
- If FQI ≥ 25, department secures full 25 marks.
- If FQI < 25, proportional marks as per calculation.

Example Filled Table

Year	X (Ph.D.)	Y (M.Tech/ME)	RF = S/20	FQI = 2.5((10X+4Y)/RF)*
CAY	15	20	40	2.5 × ((150 + 80) ÷ 40) = 14.38
CAYm1	12	18	38	2.5 × ((120 + 72) ÷ 38) = 12.63
CAYm2	10	15	36	2.5 × ((100 + 60) ÷ 36) = 11.11
Average	–	–	–	(14.38 + 12.63 + 11.11) ÷ 3 = 12.71

5.3. Faculty Cadre Proportion (25)

1. What is being assessed?

- **NBA checks whether the faculty distribution across cadres (Professor, Associate Professor, Assistant Professor) in a department matches the ideal ratio prescribed by AICTE/UGC.**
- **The ideal proportion is:**

  1:2:6

  - **1 Professor**
  - **2 Associate Professors**
  - **6 Assistant Professors**
    (for every 9 faculty required to satisfy the 20:1 student–faculty ratio).

2. Formula for Required Faculty (RF)

- **First calculate total faculty required based on student strength:**

  $RF=S/20$

  where S = total number of students in the department + allied departments (UG + PG, as per section 5.1).
- **Then split RF into cadre requirements:**
  - **RF1 (Professors required) = $1/9 \times RF$**
  - **RF2 (Associates required) = $2/9 \times RF$**
  - **RF3 (Assistants required) = $6/9 \times RF$**

3. How to Evaluate Actual Faculty vs. Required Faculty

- Count actual faculty in each cadre:
  - AF1 = Actual Professors
  - AF2 = Actual Associate Professors
  - AF3 = Actual Assistant Professors
- Compare with required values (RF1, RF2, RF3).

4. Marking Scheme (Faculty Cadre Proportion Index – FCPI)

- NBA generally uses the following formula to assign marks:

  $$FCPI=(AF1/RF1+AF2/RF2+AF3/RF3)\div 3\times 25$$

- Conditions:
  - Each ratio is capped at 1.0 (i.e., excess faculty in one cadre cannot compensate for shortage in another).
  - Maximum marks = 25.

5. Step-by-Step Example

Suppose a department has S = 720 students.

1. Compute RF (Required Faculty):

   $$RF=720/20=36$$

2. Distribute as per 1:2:6:
   - RF1 (Professors) = 36 × (1/9) = 4
   - RF2 (Associates) = 36 × (2/9) = 8
   - RF3 (Assistants) = 36 × (6/9) = 24
3. Actual Faculty (assume):
   - AF1 = 3 Professors
   - AF2 = 6 Associates
   - AF3 = 28 Assistants
4. Compute ratios:
   - AF1/RF1 = 3/4 = 0.75
   - AF2/RF2 = 6/8 = 0.75
   - AF3/RF3 = 28/24 = 1.16 → capped at 1.0

5.Compute FCPI:

$$FCPI=(0.75+0.75+1.0)/3\times25=20.83$$

Marks obtained = 20.83 / 25

6. Key Notes for Compliance

- Faculty qualifications & experience must meet AICTE/UGC norms for cadre designation.
  - Professor = Ph.D. + minimum experience.
  - Associate Professor = Ph.D./M.Tech + prescribed experience.
  - Assistant Professor = M.Tech/ME or equivalent.
- Exclusions: Do not count visiting/adjunct/part-time faculty.
- Allied departments: Their faculty must be included if their students are counted in S.

7. Documentation Required

- Faculty roster with designation, qualification, experience, and approvals (AICTE/UGC).
- Student strength (S) calculations as per Criterion 5.1.
- Appointment letters, approvals, and payroll records to validate full-time status.

Faculty Cadre Proportion (25 marks) with your given table format and formula

1. Objective

- NBA evaluates whether a department maintains the right balance of senior and junior faculty.
- The required distribution is set by AICTE/UGC norms and is tested against Student–Faculty Ratio (SFR) from section 5.1.
- Marks are awarded based on how close actual cadre availability (AF1, AF2, AF3) matches the required (RF1, RF2, RF3).

2. Cadre Definitions

- RF1 / AF1 = Professors
- RF2 / AF2 = Associate Professors
- RF3 / AF3 = Assistant Professors

Required faculty numbers (RF) are computed from student strength (S):

$RF=S/20$

Then distributed as:

- $RF1 = (1/9) \times RF$ (Professors)
- $RF2 = (2/9) \times RF$ (Associates)
- $RF3 = (6/9) \times RF$ (Assistants)

3. Marks Formula

Faculty Cadre Proportion Marks=(AF1/RF1+0.6AF2/RF2+0.4AF3/RF3)×12.5

- **AF1/RF1 = fulfillment ratio for Professors**
- **AF2/RF2 = fulfillment ratio for Associates**
- **AF3/RF3 = fulfillment ratio for Assistants**
- **Ratios can be greater than 1 if extra faculty are available.**
- **If AF1 = 0 and AF2 = 0, marks = 0 (department lacks senior leadership).**
- **Maximum = 25 marks (cap applied).**

4. Example Cases

- **Case 1: Perfect requirement met**
  - **AF1/RF1 = 1, AF2/RF2 = 1, AF3/RF3 = 1**
  - **Marks = (1 + 0.6 + 0.4) × 12.5 = 25**
- **Case 2: Extra Associates, shortage in Assistants**
  - **AF1/RF1 = 1, AF2/RF2 = 4/2 = 2, AF3/RF3 = 8/9 ≈ 0.89**
  - **Marks = (1 + 1.2 + 0.36) × 12.5 = 32 → limited to 25**

Table Structure (Assessment over 3 years)

Year	RF1 (Professors Req.)	AF1 (Available)	RF2 (Associates Req.)	AF2 (Available)	RF3 (Assistants Req.)	AF3 (Available)	Marks (Capped at 25)
CAY							
CAYm1							
CAYm2							
Average	–	–	–	–	–	–	AVG(3 years)

. **Key Notes**

- **Inclusions:**
  - **Faculty from department + allied departments (if their students are counted in S).**
  - **Only AICTE/UGC approved, full-time faculty.**
- **Exclusions:**
  - **MCA, BCA, MBA, and other non-engineering programs (their faculty and students are not counted in Table 5.3.1).**
  - **Visiting/adjunct/part-time faculty.**
  - **100% first-year faculty (already excluded in SFR).**

5.4. Visiting/Adjunct Faculty/Professor of Practice (10)

1. Objective of this Criterion

- NBA wants to see active engagement with external experts in teaching and mentoring students.
- These can be:
  - Visiting faculty (from other universities or institutions).
  - Adjunct faculty (professionals appointed part-time to teach specific subjects).
  - Professors of Practice (senior professionals from industry/research bodies engaged for teaching).
  - Emeritus/Retired professors (continuing to contribute in teaching and learning).

2. Marks Distribution (Maximum = 10 Marks)

1. Provision (1 mark)
   - If the institute has a formal system of appointing visiting/adjunct/professors of practice.
   - Evidence: Appointment letters, MoUs, Governing body approval, policy documents.
2. Interaction Hours (9 marks)
   - Requirement: Minimum 50 hours per year of interaction with adjunct/visiting faculty from industry, research organizations, or retired professors.
   - Scoring Rule:
     - Each year with ≥50 hours = 3 marks.
     - For 3 years (CAY, CAYm1, CAYm2), maximum = 9 marks.

Total maximum = 1 + 9 = 10 marks.

3. What Counts as “Interaction”

- Teaching courses/lectures (theory classes, guest lectures, seminars).
- Laboratory/practical sessions.
- Workshops/short-term training programs.
- Mentorship for student projects, startups, or research.
- Industry-oriented training modules (internship guidance, employability skills).

4. Step-by-Step Evaluation

1. Check if visiting/adjunct faculty system exists → award 1 mark.
2. Collect year-wise records of contributions:
    - CAY: Hours contributed (≥50 = 3 marks).
    - CAYm1: Hours contributed (≥50 = 3 marks).
    - CAYm2: Hours contributed (≥50 = 3 marks).
3. Calculate total marks:
    - Provision (1 mark) + Sum of yearly interaction marks (up to 9).
    - Cap at 10 marks maximum.

Example Table (Assessment)

Year	Provision of Visiting/Adjunct Faculty (Y/N)	Hours of Interaction	Marks Awarded (max 3/year)
CAY	Yes	65	3
CAYm1	Yes	50	3
CAYm2	Yes	72	3
Provision Marks	–	–	1
Total Marks	–	–	10

Key Notes

- **Quality check: NBA team will check whether contributions were substantive (not just symbolic guest lectures).**
- **Documentation required:**
  - **Appointment letters / MoUs / contracts with adjunct/visiting faculty.**
  - **Timetable records showing hours delivered.**
  - **Attendance sheets / feedback forms of guest lectures.**
  - **Report of contribution (course taught, projects mentored, workshops conducted).**
- **Sustainability: The practice should be continuous across the 3 assessment years.**

List of Visiting/Adjunct Faculty/Professors of Practice and their teaching/practical loads

1. Purpose

- NBA requires proof that external experts (visiting, adjunct, emeritus, professors of practice) are actively contributing to teaching and practical sessions.
- The course name and hours handled must be tracked year-wise (CAY, CAYm1, CAYm2).
- Helps evaluate compliance with the ≥50 hours/year requirement for awarding marks.

2. Template Structure

S.N.	Name of the Person	Designation & Organization	Name of the Course / Activity	No. of Hours Handled
1	Prof. A.B. Expert	Senior Scientist, DRDO	Cyber Security (Elective)	22
2	Dr. C.D. Professional	Adjunct, Infosys Ltd.	Data Science Lab + Workshop	35
Total No. of Hours:	–	–	–	57

3. Year-wise Format (as per your request)

CAYm1

S.N.	Name of the Person	Designation & Organization	Name of the Course	No. of Hours Handled
1				
2				
Total No. of Hours:	–	–	–	…

CAYm2

(same format, new entries)

CAYm3

(same format, new entries)

4. NBA Evaluation Logic

- If total ≥50 hours in a year → that year earns 3 marks.
- Must document each faculty's contribution with:
  - Appointment/MoU letter
  - Course allocation / timetable proof
  - Attendance sheet / feedback forms
  - Report of hours delivered

5. How to Use

- Maintain one sheet per year (CAY, CAYm1, CAYm2).
- Record each faculty separately with their hours.
- At the bottom, total hours → used in Assessment Summary to award marks.

5.5. Faculty Retention (10)

1. Objective of this Criterion

- NBA wants to measure the stability of faculty in the department.
- Frequent turnover of faculty affects continuity in teaching, mentoring, and research.
- Retention ensures a strong academic ecosystem and reduces reliance on temporary/contract staff.

2. How Retention is Measured

- Retention = % of faculty members who have served in the department for 3 or more years (continuously).
- Faculty considered:
  - Only full-time, regular/approved faculty (counted in SFR).
  - Exclude visiting/adjunct/part-time and first-year-only faculty.

3. Formula

Retention %=Number of faculty retained for ≥ 3 years/Total faculty required as per SFR (RF)×100

4. Marks Distribution (Maximum = 10)

- Retention ≥ 90% → 10 marks
- Retention ≥ 75% but < 90% → 8 marks
- Retention ≥ 60% but < 75% → 6 marks
- Retention ≥ 50% but < 60% → 4 marks
- Retention < 50% → 0 marks

5. Step-by-Step Process

1. Calculate RF (Required Faculty):
   - From Criterion 5.1 → RF = S / 20 (total students / 20).
2. Identify faculty retained for ≥ 3 years:
   - Cross-check appointment date → still employed in CAY.
   - Collect supporting evidence (joining report, payroll continuity).
3. Compute Retention %:
   - Divide faculty retained ≥3 years by RF.
4. Award marks based on NBA bands above.

6. Example Calculation

- RF (Required Faculty): 40
- Actual Retained Faculty (≥ 3 years): 34

Retention % = (34 ÷ 40) × 100 = 85%

Marks = 8/10 (falls in 75–90% band).

7. Documentation Required

- Faculty retention register with:
  - Faculty name, designation, date of joining, status (retained or not).
- Joining reports + payroll continuity records for verification.
- AICTE/University approvals to prove faculty is regular.

8. Importance for NBA

- High retention = continuity, experienced faculty pool, better outcomes in student mentoring, R&D, accreditation cycles.
- Low retention = signals instability → negative impact on marks

Faculty Retention Ratio Calculator

1. Objective

- NBA evaluates not just how many faculty are retained, but also their length of service in the same institute.
- A stable faculty team ensures continuity in teaching, mentoring, and research.
- Retention is converted into a quantitative score (Faculty Retention Ratio – FR).

2. Key Definitions

- RF = Required Faculty = S/20
  - Where S = total students in the department + allied departments (from section 5.1).
  - Exclude MCA/BCA and other non-engineering programs.
- AF = Available Faculty (in the Department + Allied Departments, counted in SFR).
- Faculty categorization (experience at the current institute):
  - A = Faculty with < 1 year experience (not retained yet).
  - B = Faculty with ≥1 year and <2 years.
  - C = Faculty with ≥2 years and <3 years.
  - D = Faculty with ≥3 years and <4 years.
  - E = Faculty with ≥4 years.

3. Formula for Faculty Retention (per year)

$FR=((A\times0)+(B\times1)+(C\times2)+(D\times3)+(E\times4)/RF)\times2.50$

- The score is capped at 10 points per year.
- This formula gives higher weightage to longer-serving faculty:
  - 0 for <1 year,
  - 1 for 1–2 years,
  - 2 for 2–3 years,
  - 3 for 3–4 years,
  - 4 for ≥4 years.

4. Average Faculty Retention Score

$Average\ FR=(FR1+FR2+FR3)/3$

- Computed across CAYm1, CAYm2, and CAYm3.
- Limited to 10 marks maximum.

5. Example Calculation

Suppose in CAYm1:

- RF = 40
- Faculty distribution: A=3, B=4, C=5, D=8, E=20

$FR=((3\times0)+(4\times1)+(5\times2)+(8\times3)+(20\times4)/40)\times2.5$

$FR=(0+4+10+24+80/40)\times2.5$

$FR=(118/40)\times2.5=7.375$

Marks for CAYm1 = 7.38/10

Similarly, calculate for CAYm2 and CAYm3, then average.

Faculty Retention Ratio Table

Item	CAYm1	CAYm2	CAYm3
RF (Required Faculty = S/20)			
AF (Available Faculty)			
A (<1 year)			
B (≥1 & <2 years)			
C (≥2 & <3 years)			
D (≥3 & <4 years)			
E (≥4 years)			
Check: A+B+C+D+E			
Mismatch? (AF – Sum(A..E))			
Weighted = (0·A + 1·B + 2·C + 3·D + 4·E)			
FR (per year) = ((Weighted / RF) × 2.5), max 10			
Average FR = $(FR_1 + FR_2 + FR_3)/3$, max 10	→	→	Final

Notes (as per NBA guidelines)

- **RF must include students + allied departments (from Criterion 5.1 SFR).**
- **Exclude MCA, BCA, MBA, and other non-engineering programs (both students and faculty).**
- **Faculty evidence required:**
  - **Appointment letters + joining reports.**
  - **Service records/payroll continuity.**
  - **Faculty approval letters (AICTE/University).**

1. Required Faculty (RF = S/20)

- CAYm1: 2,357 ÷ 20 = 117
- CAYm2: 2,345 ÷ 20 = 117
- CAYm3: 2,307 ÷ 20 = 115

2. Available Faculty (AF)

- CAYm1: 120
- CAYm2: 124
- CAYm3: 125

3. Faculty Distribution (Experience in same Institute)

Category	Definition	CAYm1	CAYm2	CAYm3
A	<1 year	1	0	2
B	≥1 & <2 years	2	3	10
C	≥2 & <3 years	1	2	2
D	≥3 & <4 years	1	0	1
E	≥4 years	115	119	110

4. Weighted Calculation

Formula:

Weighted=(0×A)+(1×B)+(2×C)+(3×D)+(4×E)

- **CAYm1: (0×1) + (1×2) + (2×1) + (3×1) + (4×115)**
  = 0 + 2 + 2 + 3 + 460 = 467
- **CAYm2: (0×0) + (1×3) + (2×2) + (3×0) + (4×119)**
  = 0 + 3 + 4 + 0 + 476 = 483
- **CAYm3: (0×2) + (1×10) + (2×2) + (3×1) + (4×110)**
  = 0 + 10 + 4 + 3 + 440 = 457

5. FR (per year)

FR=(Weighted/RF)×2.5(max 10)

- **CAYm1: (467 ÷ 117) × 2.5 = 9.98**
- **CAYm2: (483 ÷ 117) × 2.5 = 10.32 → capped at 10.00**
- **CAYm3: (457 ÷ 115) × 2.5 = 9.93**

6. Average FR (Final Marks)

AverageFR=(FR1+FR2+FR3)/3

= (9.98 + 10.00 + 9.93) ÷ 3

= 29.91 ÷ 3 = 9.97

Final Faculty Retention Ratio Marks = 9.97/10